Lecture 8/27/25

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Space of Functions
Input = Training
Output = Neural Network
Because we have many hyperparameters, we can make many different combinations that will result with us getting many different Neural Networks

Hyperparameter tuning is poorly understood -> Interesting

Existing Approaches to Hyperparameter tuning
- Manual Tuning, Grid search, random search:
  - Not very good, inefficient
  - Not optimal

State of the Art approaches:
- Bayesian Optimization
  - Leaps and bounds better than grid search, randomized search, manual tuning
  - Uses a confidence interval to guess the value of the true function
  - Avoid overfitting by using training, validation, and testing data
  - Guarantees need strong prior assumptions
  - Need design of kernels and acquisition functions
- Gradient-based:
  - Back Propagation?
  - Used to train model parameters
  - Can be very slow
  - Global optimality needs unrealistic smoothness assumptions
- Bandit-based:
  - One armed bandit = Slot Machine
    - We don't know the underlying probabilities
    - When should we stop pulling the lever?
    - Multi-Armed Bandit problem:
      - Each handle is a different hyperparameter configuration
      - Each time you pull a handle, you are doing training. Pay the cost
      - Hyperband: noisy non-stationary reward that converges to a value

All approaches are black box
- Can be used with any structure

Don't do it manually! Use automatic tuning methods

Exploration vs Exploitation
- Cost vs performance
- Exploration is more time looking for the best value
- Exploitation is taking what you have and stopping looking for better
- It's important to find a good balance between the two